SULIT

Fakulti Komputeran

UNIVERSITI TEKNOLOGI MALAYSIA
MIDTERM EXAMINATION SEMESTER I 2023 / 2024

SUBJECT CODE	**: SECR2043**
SUBJECT NAME	**: OPERATING SYSTEMS**
SECTION	**: 01, 02, 05, MJIT**
TIME	**: 08:00 to 10:00 PM**
DATE/DAY	**: 12 DECEMBER 2023 (Tuesday)**
VENUES	**: Bilik Kuliah 1 (BK 1), Level 1, N28**

INSTRUCTIONS:

Answer **ALL** objective questions in the answer sheet. This test will contribute **25%** towards the total marks of 100%.

(Please Write Your Lecturer Name and Section)

Name	
Matric No.	
Year / Course	
Section	
Lecturer Name	

This question booklet consists of 8 pages including the front page.

PART A: OBJECTIVES [10 MARKS]

Answer ALL questions.

Turnaround time = completion time - arrival time

1. Which of the following are true regarding CPU scheduling criteria and algorithms?
   I. Turnaround time is the total time taken from submission to completion of a process.
   II. Response time is the time it takes for a process to start execution after it has been scheduled. first response
   III. The Aging technique is used in Round Robin scheduling to prevent starvation. priority scheduling
   IV. In the Multilevel Queue scheduling algorithm, a process can move between queues.

   Multilevel Feedback Queue 才可以

A I, II, and III
B II, III, and IV
C I and II
D All of the above

2. Consider the following statements about synchronization mechanisms:
   I. Starvation can be avoided by ensuring fairness in resource allocation. one thread at a time
   II. Mutex locks allow multiple threads to access a shared resource simultaneously.
   III. Semaphores can be binary or counting types.
   IV. The Producer-Consumer problem demonstrates challenges in managing buffer resources.

A I, III, and IV
B II and IV
C I and III
D All of the above

low priority processes may never execute

3. Which scheduling algorithm can lead to starvation?
A Round Robin
B First-Come, First-Served (FCFS)
C Shortest Job First (SJF)
D Priority Scheduling

4. Which of the following is a characteristic of deadlock?
A Processes can continue execution with limited resources
B Only one process is involved
C Processes are waiting indefinitely for resources held by each other
D It is easily resolved by preempting a process

5. In a Round Robin scheduling algorithm, what does the term "quantum" refer to?
A The total time for all processes
B The fixed time assigned to each process
C The maximum allowable process execution time
D The time after which priority is recalculated

6. Which scheduling algorithm is considered optimal in terms of minimizing the average waiting time for a given set of processes?

A Round Robin

B Priority Scheduling

C Shortest Job First (SJF)

D First-Come, First-Served (FCFS)

7. Which of the following best describes a semaphore initialized to 1?

A A counting semaphore

B A binary semaphore

C A spinlock

D A barrier

8. Which of the following best describes the concept of 'aging' in CPU scheduling?

A Decreasing the priority of a process over time

B Increasing the priority of a process over time

C Reducing the quantum time for a process

D Extending the execution time of a process

9. What is the primary goal of synchronization techniques in a concurrent computing environment?

A To maximize CPU utilization

B To ensure data consistency and prevent race conditions

C To increase the throughput of processes

D To reduce the complexity of process management

10. What is the purpose of the 'wait' and 'signal' operations in semaphore implementation?

A To create and destroy semaphores

B To increase and decrease the value of the semaphore

C To lock and unlock access to shared resources

D To start and stop processes

Name	
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1		**6**	
2		**7**	
3		**8**	
4		**9**	
5		**10**	

PART B: STRUCTURED QUESTIONS [55 MARKS]

Answer ALL questions.

QUESTION 1 [18 MARKS]

There are 3 processes in the memory space, namely processes X, Y, Z, and the dispatcher as shown in Table 1. These processes will be executed on a single processor with a single core. Process Y makes an I/O resource request at the 14th instruction cycle for 25 instruction cycles. 39
Process Z also makes a request for the I/O resource at the 44th instruction cycle and waits for the resource for 30 instruction cycles.

74

Table 1

Process	Addresses	
X	4000-4019	20
Y	3000-3008	9
Z	2000-2014	15
Dispatcher	100-104	5

Complete a Gantt chart by drawing the process state for all processes for 80 instruction cycles with three indicators for running, ready, and blocked state. Assume that the maximum CPU time for each process is 5 instruction cycles.

i. Draw a **Gantt chart** based on the information. [8]

ii. Which jobs has been **Successfully Executed**? [1] X and Y

iii. What is the **Turnaround Time** of **Process X**? [3] 74 - 0 = 74s

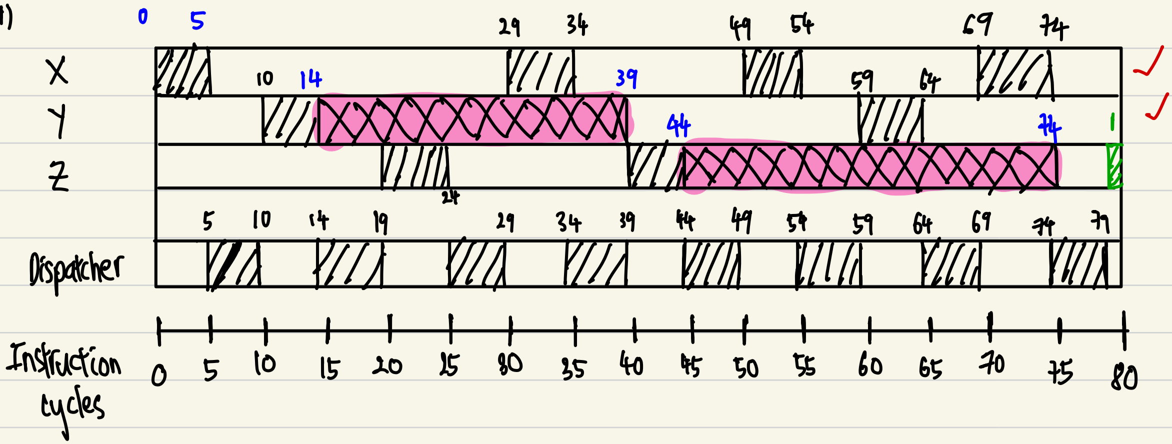
iv. What is the **Waiting Time** of **Process X**? [3] 74 - 20 = 54s

v. What is **the states** of **processes X, Y and Z** at t_{15}? [1.5] ready, blocked, ready

vi. What is **the states** of **processes X, Y and Z** at t_{50}? [1.5]

running, ready, blocked

(9)


0
5
29
34
49
54
69
74
X
10
14
39
59
64
Y
44
74
1
Z
24
5
10
14
19
29
34
39
44
49
54
59
64
69
74
79
Dispatcher
Instruction cycles
0
5
10
15
20
25
30
35
40
45
50
55
60
65
70
75
80

QUESTION 2 [18 MARKS]

Producer and Consumer

a) A fixed size buffer of 10 serves the requests between a system's Editor and Linker. At T = 0, both processes share a common variable named "Index", having an initial value = 7. Complete Table 2 (i - viii) by updating the value of Index and Event. [4]

Table 2

Time	Editor	Index	Linker	Event
T0		7		Index is initialized at 7
T1	1	(i) 8		Index is incremented by 1
T2		(ii) 7	1	Index is decremented by 1
T3	1	(iii) 8		Index is incremented by 1
T4	1	(iv) 9		Index is incremented by 1

b) Two processes are concurrently accessing their critical sections.

i. Using Peterson's Algorithm, complete Table 3 (a - m). [5.5]

Table 3

P1 requests to enter CS1

Time	flag[0]	flag[1]	Turn	Events
t0	False	True	(a) 0	(b)
t1	False	(c) True	0	P1 enters CS1
t2	True	True	1	(d) P0 requests to enter CS0.
t3	True	False	(e) 1	(f) P1 executes RS1
t4	True	True	(g) 0	(h) P1 requests to enter CS1
t5	False	True	(j) 0	(k) P0 executes RS0
t6	True	False	(l) 1	(m) P0 requests to enter CS0.

ii. From Question (i), when is the bounder waiting preserved? [1]

c) Based on the following statements, write out the possible value of semaphores and outputs for Table 4. Assume the following order sequence is executed between Process X and Y under the pre-emptive scheduler. Any process continues until one of the semaphore values turns into 0. The initial value of semaphores for Process X (R) and Process Y (S) is set to be 4 and 3, respectively. [7.5]

4 3

//Process X	//Process Y
While (1) {	While (1) {
wait (R); 3	wait (S); 2 2
printf (“Hello there. \n”);	printf (“Goodbye. \n”);
signal (R); 4	signal (S);
}	}

Table 4

(copy Table 4 to the answer booklet provided)

Process	R	S	Output
Initial	4	3	-
Process Y	4	3	Goodbye
Process X	4	3	Hello there.
Process Y	4	3	Goodbye
Process X	4	3	Hello there
Process Y	4	3	Goodbye

QUESTION 3 [19 MARKS]

Five processes, **A, B, C, D**, and **E**, arrive at different times; with varying CPU burst lengths (in milliseconds) and priority levels, as shown in **Table 5**.

Table 5

Processes	Arrival Time	Burst Time	Priority
A	0	~~7~~ 5	10
B	2	~~8~~ 3	~~4~~
C	4	~~10~~	~~2~~
D	6	8	~~5~~
E	8	~~6~~	~~3~~

a) Draw **Gantt chart** of the scheduling if

FCFS i. First Come First Serve algorithm is applied. [2.5]

SRTF ii. Shortest Remaining Time First algorithm is applied. [2.5]

RR iii. Round Robin algorithm is applied, with Time Quantum = 4. [2.5]

iv. Preemptive Priority algorithm is applied. [2.5]

b) Calculate **completion time**, **turnaround time** and **waiting time** for each process in table format if

i. First Come First Serve algorithm is applied. [1.5]

ii. Shortest Remaining Time First algorithm is applied. [1.5]

iii. Round Robin algorithm is applied, with Time Quantum = 4. [1.5]

iv. Preemptive Priority algorithm is applied. [1.5]

c) Calculate **average of turnaround time** and **average of waiting time** if

i. First Come First Serve algorithm is applied. [0.5]

ii. Shortest Remaining Time First algorithm is applied. [0.5]

iii. Preemptive Priority algorithm is applied. [0.5]

iv. Round Robin algorithm is applied, with Time Quantum = 4. [0.5]

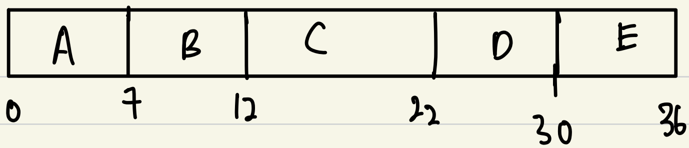
d) Which scheduling is the best scheduling algorithm? [1]

SRTF

FCFS

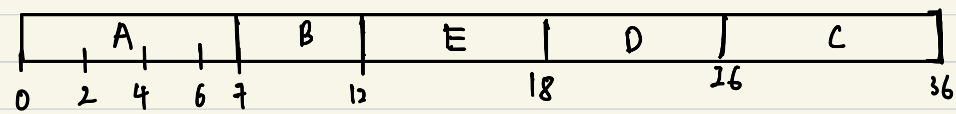
3-(a)

(i)



	Arrival time	Burst	Completion	Turnaround	Waiting
A	0	7	7	7	0
B	2	5	12	10	5
C	4	10	22	18	8
D	6	8	30	24	16
E	8	6	36	28	22
				17.4	10.2

(b) SRTF



A_7 A_5 A_3 B_5 C_{10}

B_5 C_{10} D_8

D_8 E_6

	Arrival time	Burst	Completion	Turnaround	Waiting
A	0	7	7	7	0
B	2	5	12	10	5
C	4	10	36	32	22
D	6	8	26	20	12
E	8	6	18	10	4
				15.8	8.6

(c) RR, Q=4

A	B	C	D	E	A	B	C	D	E	C

0 4 8 12 16 20 23 24 28 32 34 36

	Arrival time	Burst	Completion	Turnaround	Waiting
A	0	7	23	23	16
B	2	5	24	22	17
C	4	10	36	32	22
D	6	8	32	26	18
E	8	6	34	26	20
				25.8	18.6

(d)

A	B	C	E	B	D	A

0 2 4 14 20 23 31 36

	Arrival time	Burst	Completion	Turnaround	Waiting
A	0	7	36	36	29
B	2	5	23	21	16
C	4	10	14	10	0
D	6	8	31	25	17
E	8	6	20	12	6
				20.8	13.6